SYNAPSE EDGE INTELLIGENCE

Signal Horizon Telemetry Pipeline

From 10K signals/sec at the edge to hunt-ready rollups in ClickHouse. Authenticated ingest, tiered retention, and ten materialized views that keep dashboards snappy.

10K/s	5K	10	365d
SIGNAL THROUGHPUT	EVENTS PER BATCH	MATERIALIZED VIEWS	LONGEST RETENTION

INGEST PATH — EDGE TO HUNTER

STAGE 01
Edge Sensors
Synapse Pingora + Apparatus agents emit signal events, HTTP transactions, and sensor logs.

EMIT

STAGE 02
Telemetry API
POST /telemetry · JWT-auth'd · replay-protected via nonce store · Zod-validated. Batches up to 5000 events.

INGEST

STAGE 03
Retry Buffer
In-memory retry queue catches transient ClickHouse outages without dropping samples.

RESILIENCE

STAGE 04
ClickHouse
Partitioned by month, ORDER BY tenant + time. ZSTD compression (~100B/signal). TTL-driven tiered retention.

STORE

STAGE 05
Dashboards & Hunts
Horizon UI & analyst queries read pre-aggregated materialized views for sub-second response.

QUERY

INGEST CONTRACT

10K/s
TARGET SUSTAINED

Per-tenant ingest target with bloom-filter indices for fast IP & fingerprint lookups.

~100B
COMPRESSED/SIGNAL

ZSTD-compressed column store. 100M signals ≈ 10 GB raw.

5000
EVENTS/BATCH MAX

Enforced by TelemetryBatchSchema; larger batches rejected at the edge.

TIERED RETENTION — TTL-DRIVEN

blocklist_history BLOCK CHANGE LOG		365d
campaign_history CAMPAIGN SNAPSHOTS		180d
signal_events PRIMARY TIME-SERIES		90d
http_transactions RAW REQUEST TELEMETRY		30d
sensor_logs RAW SENSOR OUTPUT		30d

Retention enforced via ClickHouse TTL toDateTime(timestamp) + INTERVAL N DAY. Hot tables (short TTL) carry raw detail; warm tables (long TTL) carry summaries for historical hunts.

PRE-AGGREGATED VIEWS — 10 ROLLUPS FOR INSTANT QUERIES

signal_hourly_mv
TENANT · TYPE · HOUR
Hourly signal counts by tenant & signal type. Base for most dashboards.

ip_daily_mv
SOURCE_IP · DAY
Daily per-IP summary for dwell-time anomaly detection.

top_actors_hourly
SENSOR · HOUR · ACTOR
Ranked actors per sensor per hour — noisy-actor surfacing.

attack_trends_daily
CLASS · DAY
Classification counts over time for trend lines.

blocks_by_sensor_hourly
SENSOR · HOUR
Block volume per sensor — traffic vs. enforcement balance.

campaign_velocity_hourly
CAMPAIGN · HOUR
Rate of new members per campaign — escalation detection.

geo_distribution_daily
COUNTRY · DAY
Geo spread of signals; feeds world-map visualizations.

actor_sensor_matrix
ACTOR × SENSOR
Cross-sensor coverage per actor — confirms spread of an attacker.

fingerprint_spread_daily
FINGERPRINT · DAY
How many distinct IPs share a JA4 or TLS fingerprint.

daily_summary
TENANT · DAY
Top-level tenant totals — dashboards home page.

Raw events in. Hunt-ready rollups out.
Authenticated ingest, replay-protected, retry-buffered. TTL-driven tiered retention. Ten materialized views that keep the dashboards honest.